

Microorganisms and Disease — Complete Notes

Grade 11

Life Sciences

CAPS

What you'll learn

- The four main groups of microorganisms and how to tell them apart
- The basic structure and general characteristics of viruses, bacteria, protists and fungi
- How microorganisms cause disease, and how diseases are transmitted and prevented
- How the human body defends itself, and the four types of immunity
- How vaccines work, step by step
- The role of microorganisms in traditional and modern biotechnology
- How to write full, reasoned answers and do simple calculations on disease data

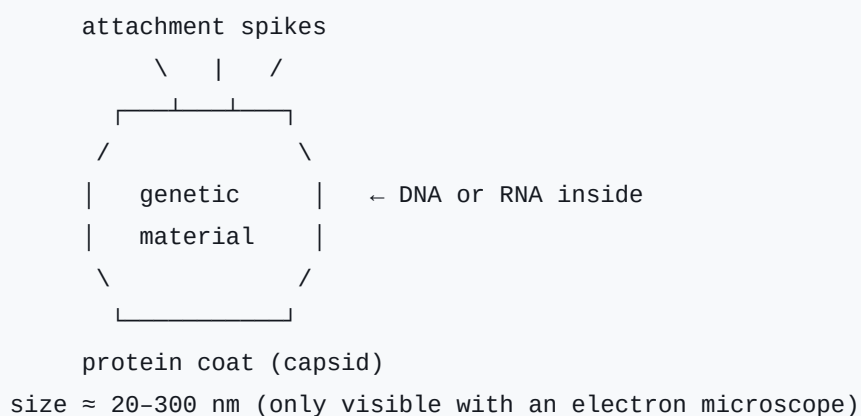
What is a microorganism?

A **microorganism** (also spelled micro-organism, or simply a **microbe**) is a living organism that is too small to be seen with the naked eye — you need a microscope. Microorganisms live almost everywhere: in soil, water and air, on your skin, and inside your body. Most are harmless or even useful, but a small number cause disease. A microorganism that causes disease is called a **pathogen**.

CAPS groups microorganisms into four main types: **viruses, bacteria, protists (protozoa) and fungi**.

Group 1: Viruses

Viruses are the odd group out. They are **not** made of cells, so they sit on the borderline between living and non-living.



General characteristics of viruses

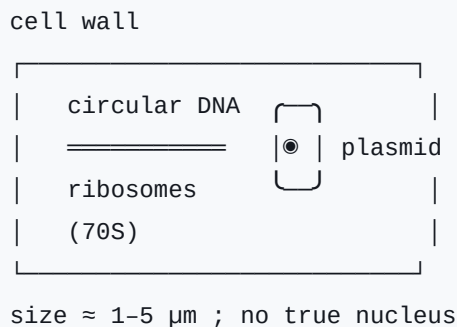
- They have **no cell structure** — no cytoplasm, no nucleus, no cell membrane and no ribosomes.
- They consist of a **protein coat (capsid)** surrounding genetic material, which is either **DNA or RNA** (never both).
- They can only **reproduce inside a living host cell** (they are obligate intracellular parasites). Outside a host they are inactive, like crystals.
- They have **no metabolism** of their own — they use the host cell's enzymes and energy.

- They are **much smaller** than bacteria (nanometres, not micrometres).
- They **mutate quickly**, which is why new strains keep appearing (e.g. new flu strains).
- Examples: HIV (AIDS), influenza, measles, rabies and the common cold viruses.

Because viruses lack the characteristics of living cells, antibiotics do **not** work against them — antibiotics only attack bacteria.

Group 2: Bacteria

Bacteria are **prokaryotes**: single-celled organisms whose genetic material is **not** enclosed in a nuclear membrane.



General characteristics of bacteria

- **Unicellular** and **prokaryotic** — there is no nuclear membrane, so the DNA lies free in the cytoplasm.
- Have a **cell wall** (mostly of a substance called peptidoglycan), a cell membrane, cytoplasm and **ribosomes**.
- The genetic material is a single **circular DNA** molecule; many bacteria also carry small rings of DNA called **plasmids**.
- Reproduce **asexually by binary fission** (one cell divides into two) — very fast, e.g. every 20 minutes in good conditions.
- Some form resistant **endospores** that survive harsh conditions.
- Nutrition: most are **heterotrophic**, but some are **autotrophic**.
- Movement: many have tail-like **flagella**.
- Many bacteria are useful (decomposers, nitrogen-fixing bacteria, those used in yoghurt and cheese), but some are pathogens: *Mycobacterium tuberculosis* (TB) and *Vibrio cholerae* (cholera).

Group 3: Protists (protozoa)

Protists are **eukaryotic** microorganisms — their DNA is enclosed in a true nucleus.

- Mostly **unicellular**, but with a true **nucleus** and membrane-bound organelles.
- **No cell wall** (in protozoa), so many can change shape.
- Move using **pseudopods, cilia or flagella**.
- Most are **heterotrophic**; some are parasites.
- Reproduce mainly by **cell division**.
- Disease-causing examples: *Plasmodium* (causes **malaria**), transmitted by the bite of an infected **female Anopheles mosquito**; also *Trypanosoma* (sleeping sickness, spread by the tsetse fly) and *Entamoeba*

(amoebic dysentery).

Group 4: Fungi

Fungi are also **eukaryotic** and can be unicellular or multicellular.

- Unicellular (e.g. **yeast**) or multicellular (e.g. **moulds**).
- Cell walls made of **chitin** (not cellulose).
- **Heterotrophic** — they digest food outside the body and absorb it.
- Reproduce **sexually or asexually by spores** — huge numbers of tiny spores are carried by air.
- They have **no chlorophyll**, so they cannot photosynthesise.
- Useful fungi: yeast in brewing and baking, *Penicillium* for antibiotics, and decomposers. Disease-causing: *Candida albicans* (thrush) and dermatophytes (ringworm and athlete's foot).

How microorganisms cause disease

Key terms

- **Pathogen** — a microorganism that causes disease.
- **Infection** — when a pathogen enters the body and multiplies.
- **Disease** — the symptoms caused by the infection.
- **Vector** — an organism that carries and transmits a pathogen from one host to another (e.g. a mosquito).
- **Carrier** — a person who carries a pathogen without showing symptoms but can still spread it.

How pathogens harm the body

- They **use the host's resources** (food, energy), so the body weakens.
- They **damage cells and tissues** (e.g. destroy lung tissue in TB).
- Many produce **toxins** (poisons) that cause fever and other symptoms.
- The body's own **immune response** also causes symptoms such as fever, swelling and pain.

Transmission and prevention

Mode of transmission	Example disease	Prevention
Airborne droplets (sneezing, coughing)	Tuberculosis, influenza	Cover mouth, ventilate rooms, vaccinate
Contaminated water or food	Cholera, typhoid	Clean water, sanitation, wash hands
Vector-borne (mosquito, tsetse fly)	Malaria, sleeping sickness	Nets, repellents, destroy breeding sites
Body fluids (blood, sex)	HIV/AIDS	Safe sex, screened blood, no shared needles
Direct contact	Ringworm, athlete's foot	Clean skin, do not share towels

Named diseases to know

Disease	Pathogen	Type of microorganism	Vector or route
Tuberculosis (TB)	<i>Mycobacterium tuberculosis</i>	Bacterium	Airborne droplets
Malaria	<i>Plasmodium</i>	Protozoan (protist)	Female <i>Anopheles</i> mosquito
HIV/AIDS	Human immunodeficiency virus	Virus	Body fluids
Cholera	<i>Vibrio cholerae</i>	Bacterium	Contaminated water
Thrush	<i>Candida albicans</i>	Fungus	Overgrowth on mucous membranes

The body's defences and immunity

The body defends itself in three levels.

- **First line of defence (non-specific, physical and chemical):** intact **skin**, **mucous membranes**, hairs and cilia in the nose, tears and saliva (which contain the enzyme lysozyme), and acid in the stomach.
- **Second line of defence (non-specific, internal):** **inflammation** (redness, heat, swelling), **phagocytes** (macrophages and neutrophils) that engulf pathogens, **fever**, and natural killer cells.
- **Third line of defence (specific):** **lymphocytes**.
- **B-lymphocytes** produce **antibodies** that bind to specific **antigens** (humoral immunity).
- **T-lymphocytes** help B-cells and destroy infected cells (cell-mediated immunity).
- Some lymphocytes become **memory cells**, so a second infection by the same pathogen is fought faster and harder.

The four types of immunity

Type	How it is acquired	Example
Natural active	Own body makes antibodies after an infection	Recovering from measles
Natural passive	Antibodies pass from mother to baby	Across the placenta; in breast milk
Artificial active	Own body makes antibodies after a vaccine	Measles or TB vaccine
Artificial passive	Antibodies are injected from another source	Anti-venom, tetanus immunoglobulin

Vaccination works by injecting a weakened, killed or harmless part of a pathogen (or its toxin). The immune system responds by making antibodies and **memory cells** without the person becoming seriously ill. If the real pathogen attacks later, the memory cells trigger a fast secondary response, so the person does not become ill.

Biotechnology

Biotechnology is the use of living organisms (or their parts) to make useful products or provide services.

- **Traditional biotechnology** — used for centuries: yeast in **bread** (CO₂ makes dough rise), **brewing** beer and wine, **Lactobacillus** bacteria in **yoghurt**, **amasi** (sour milk) and cheese, and vinegar production.

- **Modern biotechnology** — uses gene technology (genetic engineering and recombinant DNA):
- Bacteria engineered to produce human **insulin** for diabetics.
- Producing **vaccines** and **antibiotics** (penicillin from *Penicillium*).
- **Gene therapy** — inserting a normal gene into cells with a faulty one.
- Genetically modified (GM) crops with resistance to pests or drought.

Worked examples

Worked example 1 — Identify and justify

A blood sample shows a single-celled pathogen that has a true nucleus, has no cell wall, and is found inside red blood cells. The patient has recurring fevers. Name the group and suggest how it was transmitted. Justify each point.

Step 1 — Identify the group. It is a **protist (protozoan)**. The clue "single-celled organism with a true nucleus" means it is **eukaryotic**, but "no cell wall" rules out fungi.

Step 2 — Identify the likely pathogen. Living inside red blood cells and causing recurring fever points to ***Plasmodium***, the cause of **malaria**.

Step 3 — Transmission. *Plasmodium* is transmitted by the bite of an ****infected female *Anopheles* mosquito, which acts as a vector****.

Answer: Protozoan (protist); *Plasmodium*; transmitted by the *Anopheles* mosquito vector.

Worked example 2 — Explain a process (vaccination)

Explain, step by step, how a vaccine protects a person against a future infection.

Step 1 — A vaccine containing weakened, dead or harmless parts of the pathogen (**antigens**) is injected.

Step 2 — The antigen is recognised as foreign by the immune system.

Step 3 — **B-lymphocytes** produce specific **antibodies** against that antigen, and **T-lymphocytes** help.

Step 4 — Some lymphocytes become **memory cells** that remain in the body.

Step 5 — If the real, active pathogen later enters the body, the memory cells recognise it immediately and trigger a **rapid secondary immune response** — large amounts of antibody are made before symptoms develop.

Conclusion — The person is now **immune**; this is **artificial active immunity**.

Worked example 3 — Working with data

At a clinic, 1 200 people were tested for a disease. Of these, 240 tested positive. **(a)** Calculate the percentage that tested positive. **(b)** If a further 360 people were tested and **15%** were positive, how many total positive cases were there?

Step 1 (a) — Percentage positive = $(240 \div 1\,200) \times 100 = \mathbf{20\%}$.

Step 2 (b) — 15% of 360 = $(15 \div 100) \times 360 = \mathbf{54}$ additional positive cases.

Step 3 — Total positives = $240 + 54 = \mathbf{294}$.

Answer: (a) 20% (b) 294 positive cases.

Common mistakes

- Calling viruses "bacteria". Viruses are **not cells** and have **no metabolism**; bacteria are prokaryotic cells.
- Saying antibiotics cure viral diseases. Antibiotics work **only on bacteria**.
- Confusing **active** and **passive** immunity. Active means the body makes its own antibodies (lasts long); passive means ready-made antibodies are given (short-lived).
- Forgetting that a **vector** is not the pathogen — the mosquito *carries Plasmodium* but is not the parasite itself.
- Confusing a **carrier** (no symptoms but still spreads the pathogen) with a **vector** (usually another organism).
- Saying protozoa are fungi — protists are **eukaryotic**, have **no cell wall**, and belong to a separate group.
- Forgetting to justify each statement with a reason in "explain" questions.

Quick summary / cheat sheet

Group	Cells?	Nucleus?	Cell wall	Reproduces	Disease example
Viruses	No	No	No (capsid only)	Only in host cell	HIV, flu
Bacteria	Yes	No (prokaryote)	Yes	Binary fission	TB, cholera
Protists	Yes	Yes (eukaryote)	No	Cell division	Malaria
Fungi	Yes/unicellular	Yes (eukaryote)	Yes (chitin)	Spores	Thrush

- **Immunity:** active = own antibodies (long) · passive = given antibodies (short).
- **Vaccine** → **antigens** → **antibodies + memory cells** → **immunity**.
- **Biotechnology:** traditional (bread, beer, yoghurt) vs modern (insulin, vaccines, gene therapy).
- **Prevention basics:** clean water, sanitation, vaccination, vector control, hygiene.

Practice questions

1. Give **two** reasons why viruses are said to be on the borderline between living and non-living.
2. Name the microorganism group whose cell wall is made of **chitin**, and give one disease it causes.
3. Explain the difference between **natural** and **artificial** passive immunity, with one example of each.
4. A learner says, "I have flu, so I will take antibiotics." Is this correct? Explain your answer.
5. List **three** first-line defences of the human body against pathogens.
6. In a school of 800 learners, 160 have a cold. Calculate the percentage affected.
7. Explain how **binary fission** allows a bacterial infection to spread so quickly through the body.

Answers

1. They have no cell structure (no cytoplasm, nucleus or organelles); they have no metabolism of their own and can only reproduce inside a living host cell.
2. **Fungi**; for example thrush (*Candida albicans*), ringworm or athlete's foot.
3. **Natural passive** — antibodies pass from mother to baby naturally (across the placenta or in breast milk). **Artificial passive** — ready-made antibodies are injected (e.g. anti-venom or tetanus immunoglobulin).
4. Incorrect. Flu is caused by a **virus**, not a bacterium. Antibiotics only act on bacteria, so they will not kill the flu virus.

5. Any three of: intact skin; mucous membranes; cilia and hairs in the nose; tears and saliva (lysozyme); stomach acid.
6. $(160 \div 800) \times 100 = \mathbf{20\%}$.
7. In binary fission one bacterium divides into two identical cells, and each of those divides again, so the number can double roughly every 20 minutes. This exponential growth quickly produces a huge number of pathogens in the body.

Official EduRank Team notes. Always cross-check with your textbook and teacher.